

Linux Administration and a Complete Group Apache Web Project

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I completed four Linux administration assignments and worked on a shared Fedora web-server project. This walkthrough covers the individual command-line exercises and the whole team build: VM access, Apache, firewall configuration, the website, media delivery and client-side downloads.

Tools and environment

PuTTY and student VPN; Linux shell, Bash, sed, awk, systemctl, RPM, tar, cpio; VMware Workstation; Fedora 44 Server; Tailscale; Apache HTTP Server; HTML.

Individual labs: Adnan Khan. Group 9: Nawal El Khatib - VM/access; Numeir Ayaz - Apache/firewall/naming; Evan Oseen - homepage; Adnan Khan - About and Downloads pages/sample.txt; Nouf Sayeed - media tests; Furqan Haque - download confirmation and compilation. May-July 2026.

Connect and inspect the Linux environment

I connected through the student VPN and PuTTY, checked my identity with `whoami` and `id`, and used `pwd`, `ls -la`, `man` and `uname` to understand the session. I also searched configuration and log directories. Permission-denied responses marked the limits of the regular account.

```
~]$ ls -la
total 24
drwx-----  2 al0lkhan students 4096 May  1 14:20 .
drwxr-xr-x. 70 root      root    4096 May 21 16:21 ..
-rw-r--r--   1 al0lkhan students  18 Jul  7 2025 .bash_logout
-rw-r--r--   1 al0lkhan students 141 Jul  7 2025 .bash_profile
-rw-r--r--   1 al0lkhan students 376 Jul  7 2025 .bashrc
-rw-r--r--   1 al0lkhan students 172 Aug  7 2025 .kshrc
~]$
```

`ls -la`: hidden entries and long listing. Source portfolio p. 7.

```
~]$ uname -r
4.18.0-553.124.1.el8_10.x86_64
```

`uname -r`: running kernel release. Source portfolio p. 7.

Work with files, links and storage

I created and inspected files, compared a hard link with a symbolic link, and moved the original file. The hard link retained its inode relationship while the symbolic link retained the old path. I then used `df`, `du` and `/etc/fstab` to distinguish filesystem capacity, directory usage and mount configuration.

```
~]$ mkdir ~/Assignment-2
~]$ ln -s ~/MyFirstFile ~/Assignment-2/MySecondLink
~]$ ls -li ~/MyFirstFile ~/Assignment-2/MySecondLink
54157835 lrwxrwxrwx 1 al0lkhan students 26 Jun  2 20:02 /home/al0lkhan/Assignment-2/MySecondLink -> /home/al0lkhan/MyFirstFile
54157833 -rw-r--w- 2 al0lkhan students 27 May 21 16:35 /home/al0lkhan/MyFirstFile
```

Original file, hard link and symbolic link before the move. Source portfolio p. 11.

```
~]$ df -h /var/log
Filesystem              Size  Used Avail Use% Mounted on
/dev/mapper/vgroot-lvvarlog 20G  2.1G   17G  11% /var/log
```

`df -h /var/log`. Source portfolio p. 13.

Inspect memory, hardware, services and logs

I checked memory with free, swap with swapon --show and CPU information with lscpu. systemctl status sshd showed the service state; process listings connected the shell to its session. A top snapshot and the rsyslog authpriv destination added operational context.

```
~]$ free -h
              total        used        free      shared  buff/cache   available
Mem:           31Gi        1.3Gi        23Gi         32Mi         6.3Gi        29Gi
Swap:          4.1Gi          0B         4.1Gi
~]$ getconf PAGE_SIZE
4096
```

free -h and getconf PAGE_SIZE. Source portfolio p. 15.

```
~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-06-21 01:54:11 EDT; 4 days ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 1145 (sshd)
    Tasks: 1 (limit: 203852)
   Memory: 6.6M
    CGroup: /system.slice/ssh.service
            └─1145 /usr/sbin/sshd -D [REDACTED] com, [REDACTED] com,aes256-ctr,aes256-cbc,[REDACTED] com,aes128-ctr,aes128-cbc -cMACs=hmacc-sha2-256-ss

Warning: Journal has been rotated since unit was started. Log output is incomplete or unavailable.
lines 1-12/12 (END)
```

sshd status captured on the course server. Source portfolio p. 25.

Process text and practise shell control flow

I redirected normal output and errors, transformed text with sed, and filtered colon-separated records with awk. The Blue-record calculation returned Honda 35000, Ford 75000 and Nissan 30000. I also compared exported variables in child shells and wrote a three-prompt word-mapping script.

```
~]$ awk -F: '$2 == "Blue" {print $1, ($3 * $4)}' ~/assignment-3/cars
Honda 35000
Ford 75000
Nissan 30000
```

Blue records and computed totals. Source portfolio p. 20.

```
~]$ vi ~/assignment-3/opposite.sh
~]$ chmod u+x ~/assignment-3/opposite.sh
~]$ ~/assignment-3/opposite.sh
Enter a word (high/low/up/down):high
The opposite is: low
Enter a word (high/low/up/down):low
The opposite is: high
Enter a word (high/low/up/down):up
The opposite is: down
~]$ echo $?
0
```

Recorded execution and exit status. Source portfolio p. 23.

Inspect archives and installed software

I compared compress and gzip output, created tar and cpio archives, and listed their members. I then used RPM to inspect openssh-server metadata, supplied files and requirements, with ldd for runtime libraries. The archive evidence records creation and listing; a restore would be a separate validation step.

```
~]$ tar -cf ~/MytarBackup.tar ~/assignment-4
tar: Removing leading `/' from member names
~]$ ls -lh ~/MytarBackup.tar
-rw-r----- 1 al0lkhan students 50K Jul  1 17:25 /home/al0lkhan/MytarBackup.tar
~]$ tar -tvf ~/MytarBackup.tar
drwxr-x--- al0lkhan/students 0 2026-07-01 17:11 home/al0lkhan/assignment-4/
-rw-r----- al0lkhan/students 40924 2026-07-01 17:08 home/al0lkhan/assignment-4/w
tmp-2026-0209.gz
```

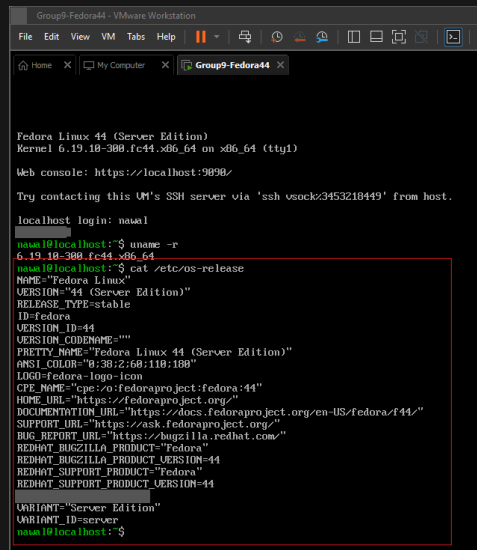
TAR creation and member listing. Source portfolio p. 28.

```
~]$ rpm -ql openssh-server
/etc/pam.d/sshd
/etc/ssh/sshd_config
/etc/sysconfig/sshd
/usr/lib/.build-id
/usr/lib/.build-id/49
/usr/lib/
/usr/lib/.build-id/d0
/usr/lib/
/usr/lib/systemd/system/sshd-keygen.target
/usr/lib/systemd/system/sshd.service
/usr/lib/systemd/system/sshd.socket
/usr/lib/tmpfiles.d/openssh.conf
/usr/libexec/openssh/sftp-server
/usr/libexec/openssh/sshd-keygen
/usr/sbin/sshd
/usr/share/man/man5/moduli.5.gz
/usr/share/man/man5/sshd_config.5.gz
/usr/share/man/man8/sftp-server.8.gz
/usr/share/man/man8/sshd.8.gz
/var/empty/sshd
```

Files supplied by openssh-server. Source portfolio p. 32.

Prepare the group VM and remote access

Our team used a Fedora 44 Server VM in VMware Workstation. Nawal prepared the base environment, bridged network access and Tailscale connectivity. The recorded addresses were 192.168.1.121 locally and 100.100.214.52 through Tailscale. SSH provided the administration path.



```
Group9-Fedora44 - VMware Workstation
File Edit View VM Tabs Help
Home My Computer Group9-Fedora44

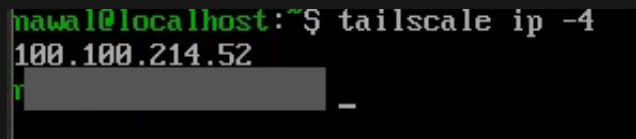
Fedora Linux 44 (Server Edition)
Kernel 6.19.18-308.fc44.x86_64 on x86_64 (tty1)

Web console: https://localhost:9090/

Try contacting this VM's SSH server via 'ssh vsock:3453218449' from host.

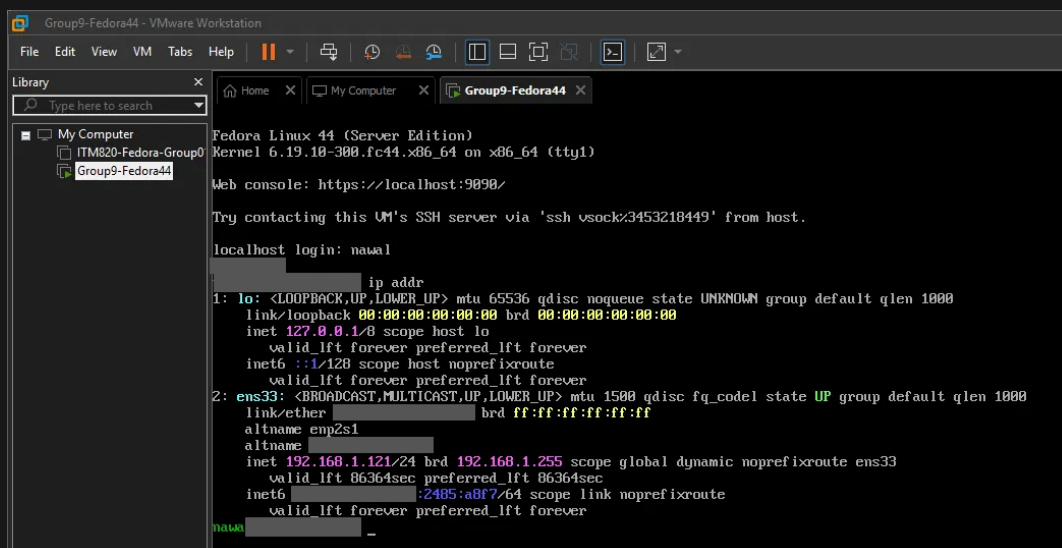
localhost login: nawal
nawal@localhost:~$ uname -r
6.19.18-308.fc44.x86_64
nawal@localhost:~$ cat /etc/os-release
NAME="Fedora Linux"
VERSION="44 (Server Edition)"
RELEASE_TYPE=stable
ID=fedora
VERSION_ID=44
VERSION_CODENAME=""
PRETTY_NAME="Fedora Linux 44 (Server Edition)"
ANSI_COLOR="0;30;2:60;110;100"
LOGO=fedora-logo-icon
CPE_NAME="cpe:/o:fedoraproject:fedora:44"
HOME_URL="https://fedoraproject.org/"
DOCUMENTATION_URL="https://docs.fedoraproject.org/en-US/fedora/f44/"
SUPPORT_URL="https://ask.fedoraproject.org/"
BUG_REPORT_URL="https://bugzilla.redhat.com/"
REDHAT_BUGZILLA_PRODUCT="Fedora"
REDHAT_BUGZILLA_PRODUCT_VERSION=44
REDHAT_SUPPORT_PRODUCT="Fedora"
REDHAT_SUPPORT_PRODUCT_VERSION=44
VARIANT="Server Edition"
VARIANT_ID=server
nawal@localhost:~$
```

Team evidence: Kernel release and Fedora 44 Server identification. Source portfolio p. 35.



```
nawal@localhost:~$ tailscale ip -4
100.100.214.52
```

Team evidence: Group VM Tailscale address. Source portfolio p. 35.



```
Group9-Fedora44 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
ITM820-Fedora-Group0
Group9-Fedora44

Fedora Linux 44 (Server Edition)
Kernel 6.19.18-308.fc44.x86_64 on x86_64 (tty1)

Web console: https://localhost:9090/

Try contacting this VM's SSH server via 'ssh vsock:3453218449' from host.

localhost login: nawal
nawal@localhost:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1f:3d:11 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname 
    inet 192.168.1.121/24 brd 192.168.1.255 scope global dynamic noprefixroute ens33
        valid_lft 86364sec preferred_lft 86364sec
    inet6 fe80::208:5a8f:764: scope link noprefixroute
        valid_lft forever preferred_lft forever
nawal@localhost:~$
```

Team evidence: VM console network-address check. Source portfolio p. 36.

```

valid ift forever preferred ift forever
sudo systemctl status sshd
[sudo] password for nawal:
• sshd.service - OpenSSH server daemon
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
  Drop-In: /usr/lib/systemd/system/service.d
           └─10-timeout-abort.conf
  Active: active (running) since Fri 2026-07-17 17:23:24 EDT; 4min 35s ago
  Invocation:
    Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 1135 (sshd)
  Tasks: 1 (limit: 2231)
  Memory: 1.6M (peak: 1.8M)
  CPU: 24ms
  CGroup: /system.slice/ssh.service
           └─1135 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 17 17:23:24 localhost.localdomain systemd[1]: Starting sshd.service - OpenSSH server daemon...
Jul 17 17:23:24 localhost.localdomain sshd[1135]: Server listening on 0.0.0.0 port 22.
Jul 17 17:23:24 localhost.localdomain sshd[1135]: Server listening on :: port 22.
Jul 17 17:23:24 localhost.localdomain systemd[1]: Started sshd.service - OpenSSH server daemon.
nawal@localhost:~$ _

```

Team evidence: sshd active on the group VM. Source portfolio p. 36.

Install Apache and check the service

Numeir installed httpd using dnf, checked its package version and enabled the service with systemctl enable --now httpd. The screenshots record httpd-2.4.68-1.fc44.x86_64 and an active Apache service. This established the web service before adding our custom content.

```

~]$ rpm -q httpd
httpd-2.4.68-1.fc44.x86_64
~]$

```

Team evidence: Installed httpd package version. Source portfolio p. 38.

```

~]$ sudo systemctl enable --now httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.

```

Team evidence: Enable and start command. Source portfolio p. 38.

```

~]$ sudo systemctl status httpd
• httpd.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
  Drop-In: /usr/lib/systemd/system/service.d
           └─10-timeout-abort.conf
  Active: active (running) since Fri 2026-07-17 19:53:29 EDT; 33s ago
  Invocation:
    Docs: man:httpd.service(8)
  Main PID: 2500 (httpd)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
  Tasks: 177 (limit: 2231)
  Memory: 14.7M (peak: 14.9M)
  CPU: 78ms
  CGroup: /system.slice/httpd.service
           └─2500 /usr/sbin/httpd -DFOREGROUND
             └─2501 /usr/sbin/httpd -DFOREGROUND
             └─2502 /usr/sbin/httpd -DFOREGROUND
             └─2504 /usr/sbin/httpd -DFOREGROUND
             └─2505 /usr/sbin/httpd -DFOREGROUND

Jul 17 19:53:29 localhost.localdomain systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jul 17 19:53:29 localhost.localdomain httpd[2500]: httpd.service: Referenced but unset environment variable evaluates to an empty string: OPTIONS
Jul 17 19:53:29 localhost.localdomain httpd[2500]: AH00559: httpd: Could not reliably determine the server's fully qualified domain name, using localhost.localdomain. Set the 'ServerName'
Jul 17 19:53:29 localhost.localdomain systemd[1]: Started httpd.service - The Apache HTTP Server.
Jul 17 19:53:29 localhost.localdomain httpd[2500]: Server configured, listening on: port 80
lines 1-24/24 (END)

```

Team evidence: Apache service status. Source portfolio p. 38.

Allow HTTP and configure the server name

The team added the HTTP firewall service permanently and reloaded the firewall. Numeir set ServerName www.group9.ca:80 in httpd.conf and restarted Apache. The directive identifies the Apache server name; the browser tests establish access within the tested client environment.

```
[redacted] ~]$ sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
success
success
[redacted] ~]$
```

Team evidence: Permanent HTTP firewall rule and reload. Source portfolio p. 39.

```
[redacted] ~]$ echo "ServerName www.group9.ca:80" | sudo tee -a /etc/httpd/conf/httpd.conf
ServerName www.group9.ca:80
[redacted] ~]$ sudo systemctl restart httpd
```

Team evidence: ServerName configuration and restart. Source portfolio p. 39.

```
[redacted] ~]$ tail -n 20 /etc/httpd/conf/httpd.conf
# Default setting is to accept 200 Ranges.
#MaxRanges unlimited

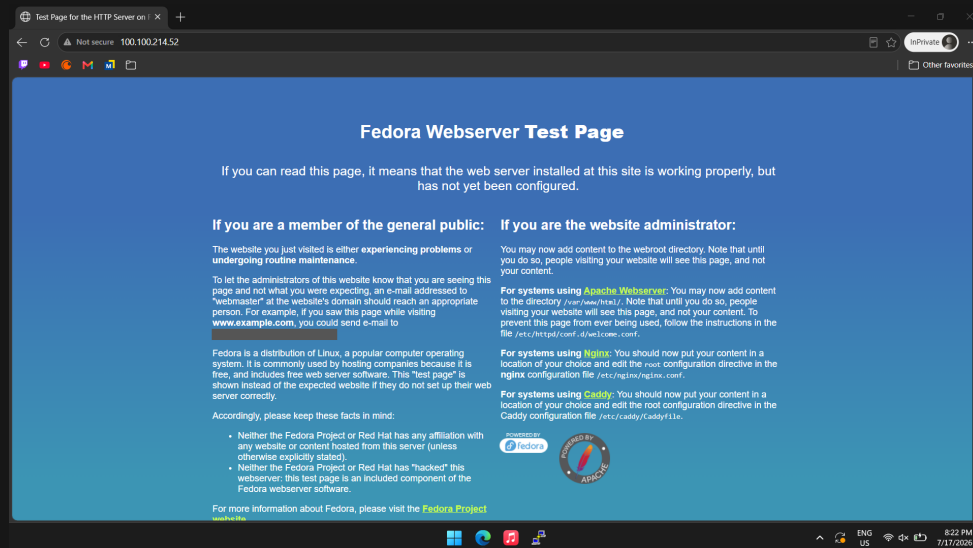
#
# EnableMMAP and EnableSendfile: On systems that support it,
# memory-mapping or the sendfile syscall may be used to deliver
# files. This usually improves server performance, but must
# be turned off when serving from networked-mounted
# filesystems or if support for these functions is otherwise
# broken on your system.
# Defaults if commented: EnableMMAP On, EnableSendfile Off
#
#EnableMMAP off
EnableSendfile on

# Supplemental configuration
#
# Load config files in the "/etc/httpd/conf.d" directory, if any.
IncludeOptional conf.d/*.conf
ServerName www.group9.ca:80
[redacted] ~]$
```

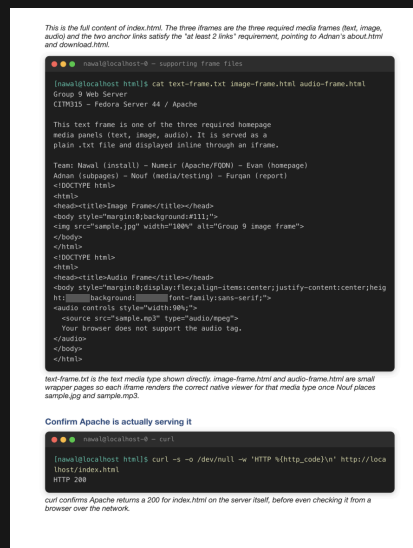
Team evidence: Servername verification. Source portfolio p. 40.

Validate the initial page and build the homepage

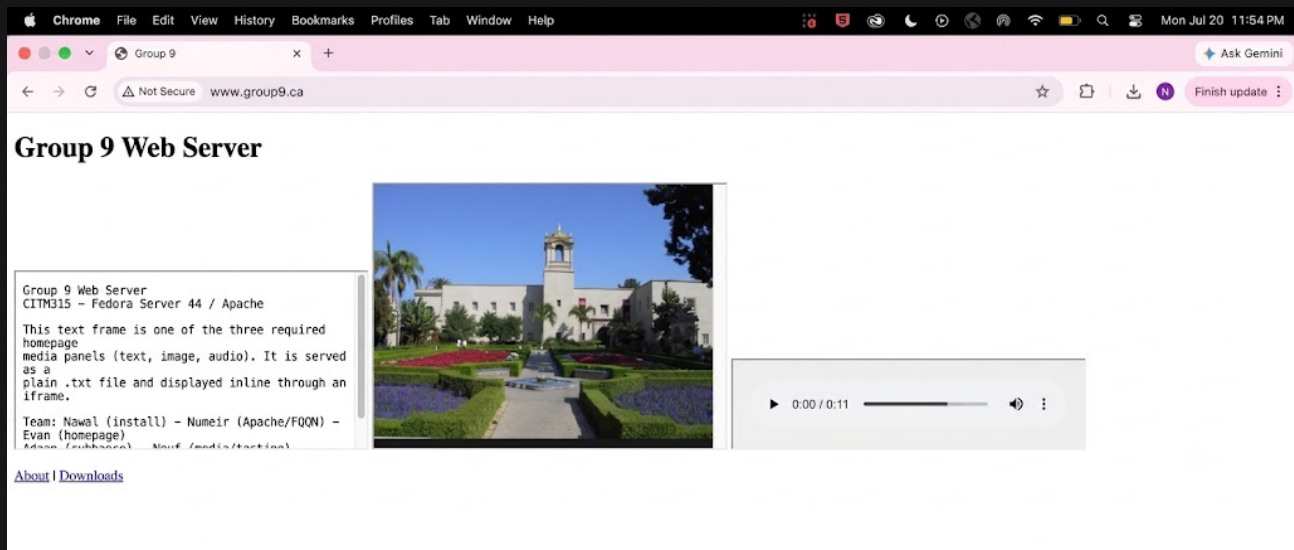
The default Fedora page was retrieved before the custom site was added. Evan built the homepage with text, image and audio content, then linked the About and Downloads pages. A curl response and the completed browser page connect the server setup to the published lab content.



Team evidence: Initial browser test. Source portfolio p. 41.



Team evidence: Homepage source and http check. Source portfolio p. 46.



Team evidence: Completed group homepage. Source portfolio p. 47.

Create the About and Downloads pages

I created about.html, download.html and sample.txt under /var/www/html. The Downloads page links to the text file using the download attribute, and both subpages link back to the homepage. I checked the rendered pages as part of the shared site.

```
~]# ls -l /var/www/html/about.html /var/www/html/download.html /var/www/html/sample.txt
-rw-r--r--. 1 root root 207 Jul 20 01:05 /var/www/html/about.html
-rw-r--r--. 1 root root 202 Jul 20 01:06 /var/www/html/download.html
-rw-r--r--. 1 root root 48 Jul 20 01:06 /var/www/html/sample.txt
~]#
```

Team evidence: about.html, download.html and sample.txt under the document root. Source portfolio p. 43.

```
~]# cat /var/www/html/about.html
<!DOCTYPE html>
<html>
<head><title>About Group 9</title></head>
<body>
<h1>About This Project</h1>
<p>Group 9 Apache web server for CITM315.</p>
<p><a href="index.html">Back to Home</a></p>
</body>
</html>
~]#
```

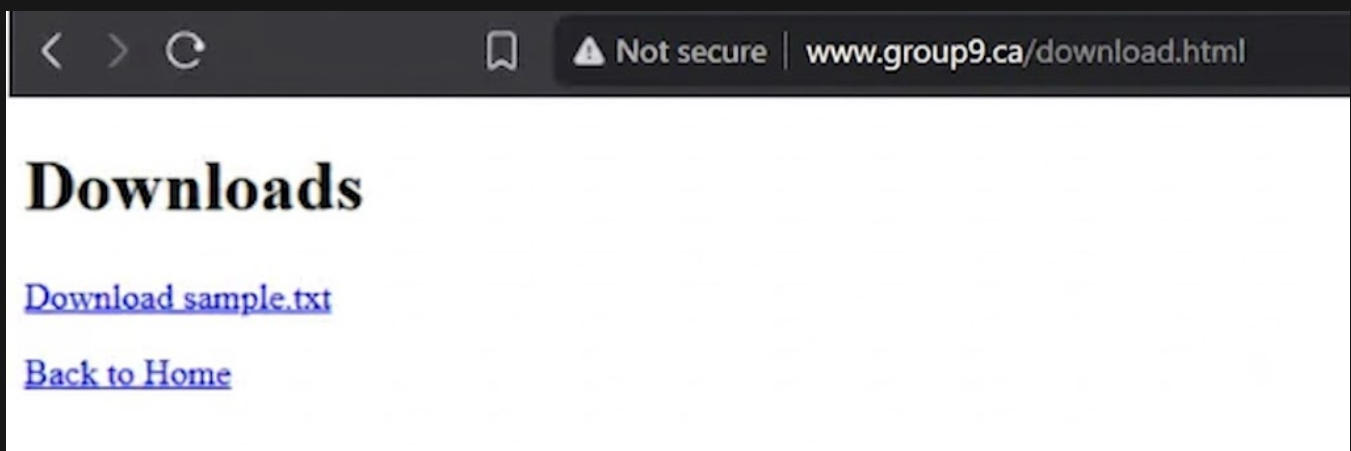
Team evidence: About page HTML. Source portfolio p. 43.

```
~]$ cat /var/www/html/download.html
<!DOCTYPE html>
<html>
<head><title>Downloads</title></head>
<body>
<h1>Downloads</h1>
<a href="sample.txt" download>Download sample.txt</a>
<p><a href="index.html">Back to Home</a></p>
</body>
</html>
~]$
```

Team evidence: Downloads page HTML with the download attribute. Source portfolio p. 43.



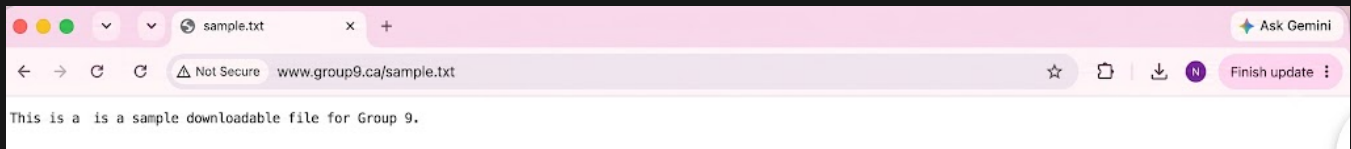
Team evidence: About page and return navigation. Source portfolio p. 44.



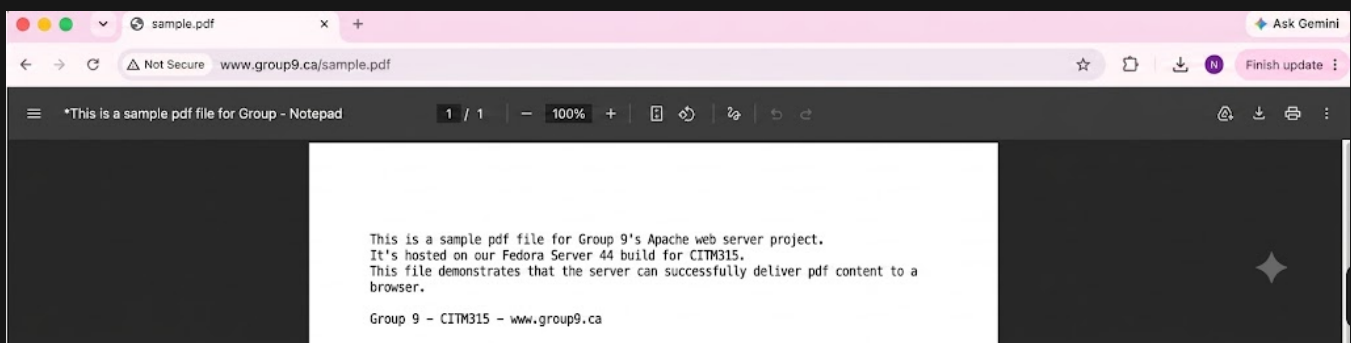
Team evidence: Downloads page and sample.txt link. Source portfolio p. 44.

Test the full range of site content

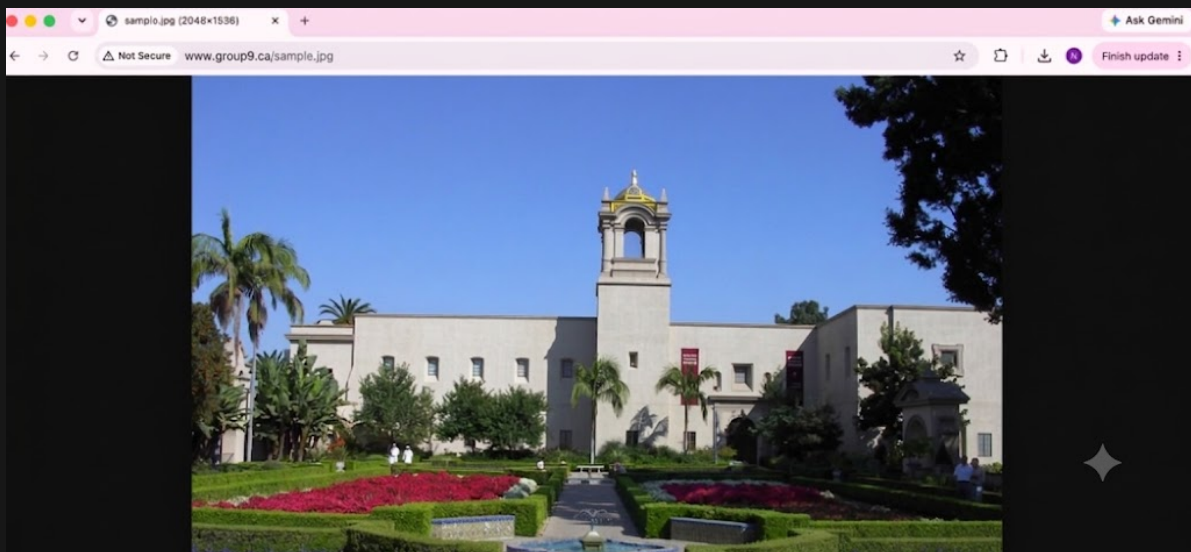
Nouf checked direct retrieval of text, PDF, image, audio and video content. These checks covered more than the homepage: each request asked Apache for a different file type under the document root. The captured browser responses show the media reached the client.



Team evidence: sample.txt displayed in the browser. Source portfolio p. 48.



Team evidence: sample.pdf rendered by the browser PDF viewer. Source portfolio p. 48.



Team evidence: Image retrieval. Source portfolio p. 49.



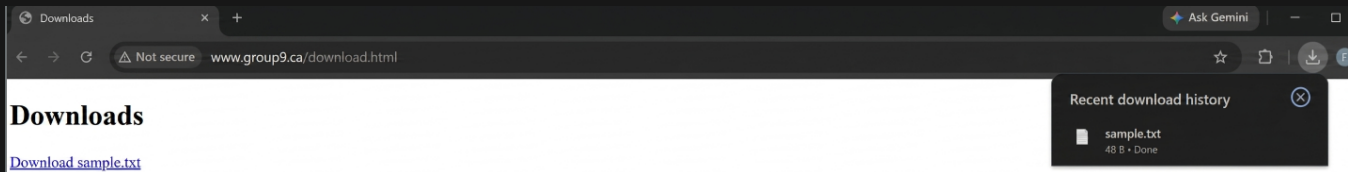
Team evidence: MP3 browser player. Source portfolio p. 50.



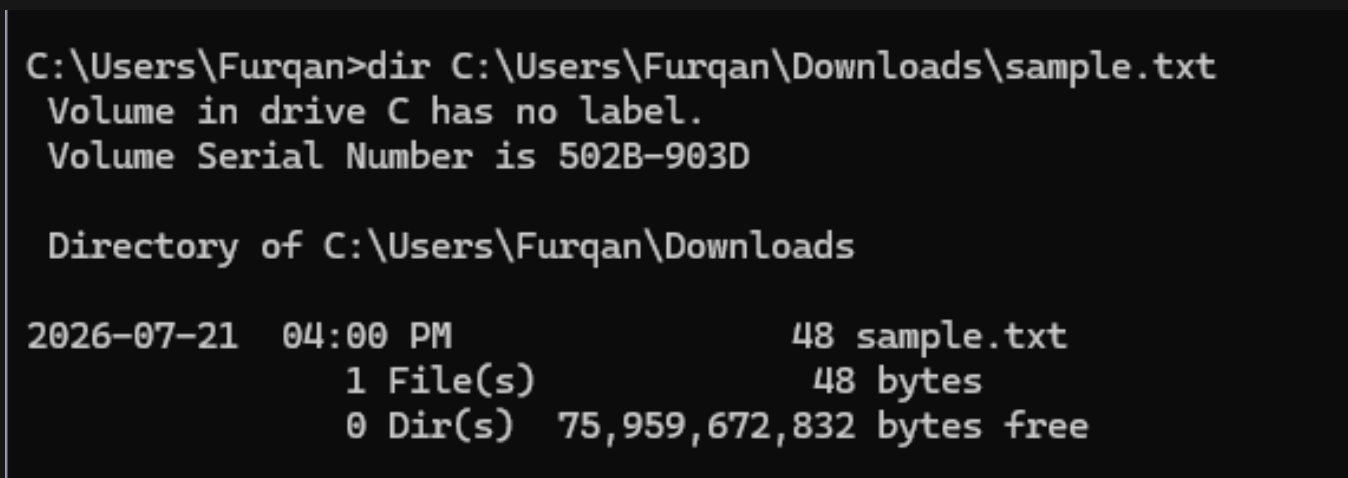
Team evidence: MP4 browser player and video frame. Source portfolio p. 50.

Confirm the download and explain the request path

Furqan confirmed the text-file download on a Windows client; the directory listing records sample.txt at 48 bytes. The complete path is browser request, Apache on the Fedora VM, content under /var/www/html, and a response or saved file on the client. SSH administration uses a separate service path.



Team evidence: Browser download history showing sample.txt completed. Source portfolio p. 51.



Team evidence: Windows directory listing showing the 48-byte downloaded file. Source portfolio p. 51.

What I learned

- Diagnose the relevant filesystem, service and package instead of relying on one command.
- Validate a website from both the server configuration and client response.
- Treat the team deployment as one system while keeping ownership of each contribution clear.

Source: the full CITM315 portfolio supplied for this project. Evidence is from the original coursework; this edition adds no new lab test results. Repeated captures and earlier drafts are consolidated. Revised design summaries are labelled, and sensitive identifiers are masked.